

Root [0]

Technical Sheet Only!
Please refer to the subsequent sheets for schematic

FPGA



File: fpga.kicad_sch

Connectivity



File: Connectivity.kicad_sch

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
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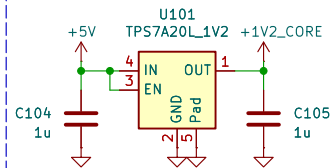
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File: Core_Module.kicad_sch

Title: Root

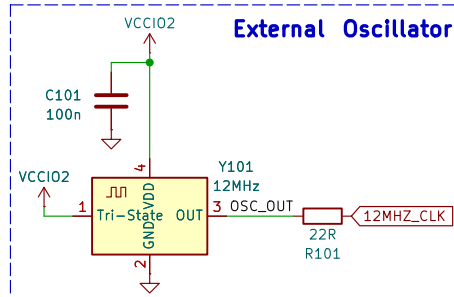
Size: A4	Date: 2026-03-31	Rev: 2.0
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FPGA [1]

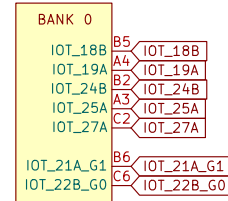
Core Supply Voltage



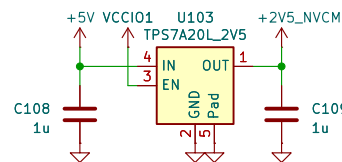
External Oscillator



Bank 0 I/O



NVCM Supply

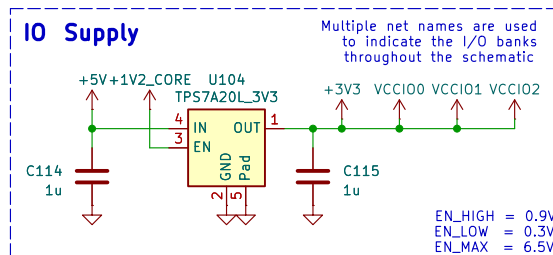


The NVCM regulator can be omitted and the VPP2V5 pin can be connected directly to 3V3 in order to save cost.

The pin can handle 3.6V.

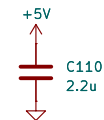
A diode can be used to drop the voltage before by roughly 0.7V

IO Supply



EN_HIGH = 0.9V
EN_LOW = 0.3V
EN_MAX = 6.5V

Input Bulk



FPGA Power Input

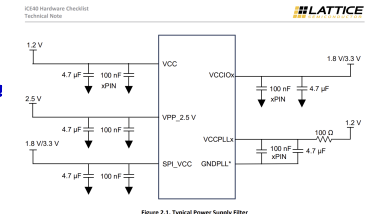
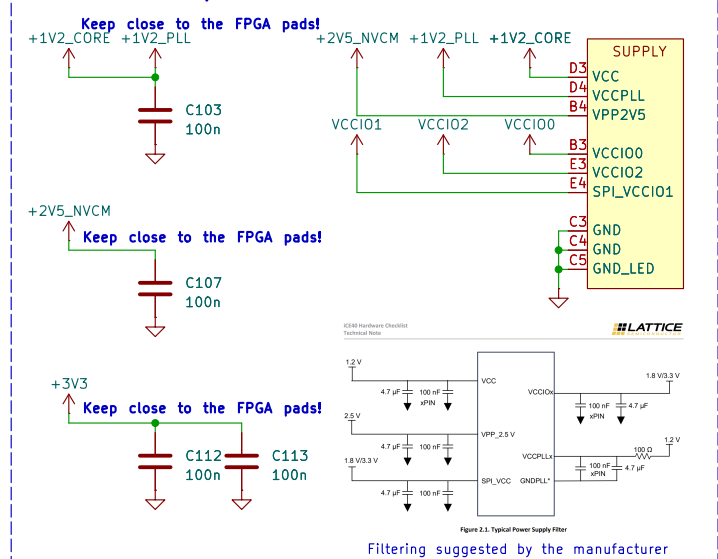
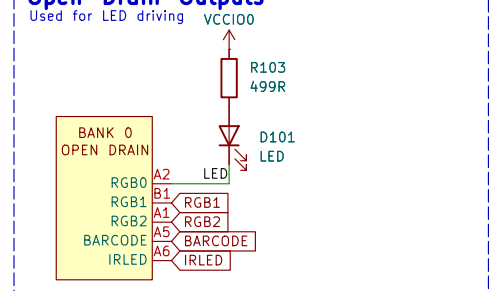
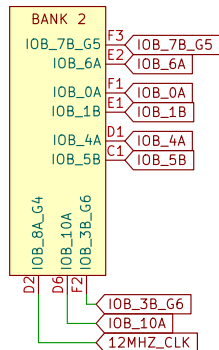


Figure 2-1: Typical Power Supply Filter

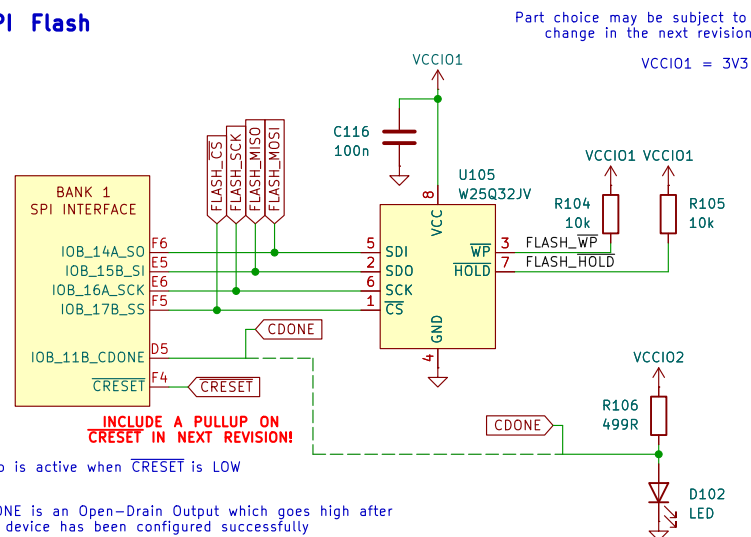
Open-Drain Outputs



Bank 2 I/O



SPI Flash



Chip is active when $\overline{\text{CRESET}}$ is LOW

CDONE is an Open-Drain Output which goes high after the device has been configured successfully

See: iCE40 Programming and Configuration app note
Section 9.2. SPI PROM Requirements

If the LED shines, the device has been configured properly

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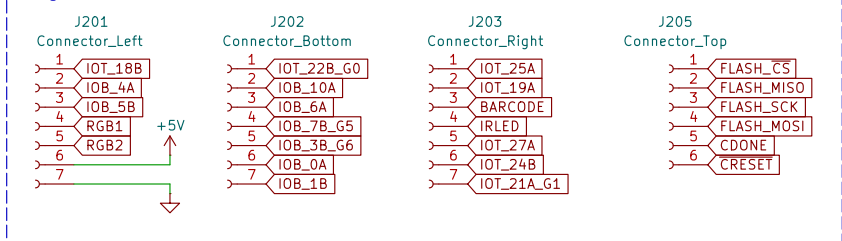
Title: FPGA

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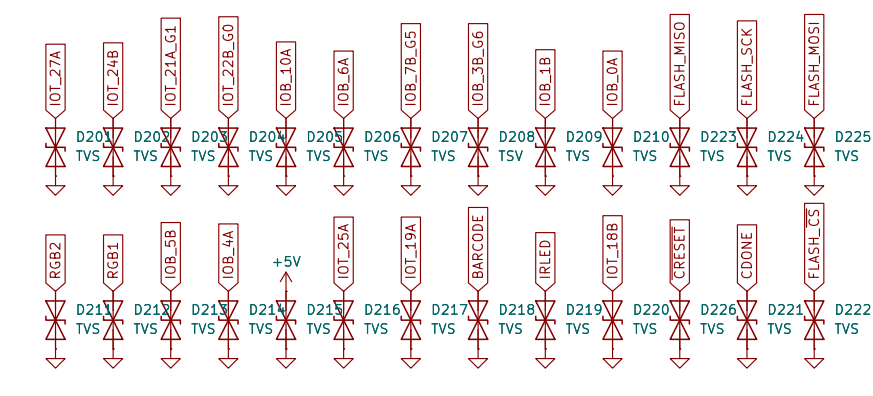
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Connectivity [2]

Edge Connectors



ESD Protection



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Title: Connectivity

Size: A4 Date: 2026-03-31
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Rev: 2.0
Id: 2/3